

# ROBIN SINGH TOMAR

## ROBOTICIST

### CONTACT

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 [Linkedin Profile](#)

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### SKILLS

|       |                                                                                     |
|-------|-------------------------------------------------------------------------------------|
| ROS   |    |
| ROS 2 |    |
| C++   |    |
| Linux |  |
| Git   |  |

### EXPERTISE

- **Problem solving.**
- **Communication protocols:** RS485, RS232, USB, CAN, **EtherCAT**, Modbus, Ethernet and other networking protocols.
- **Robot navigation and control:** Velocity Profiling, Path planning for single and multi agents, **Trajectory planning** and smoothing.
- **Robot localisation:** Barcode based, Natural feature based.
- **WiFi communication tuning,** debugging and enhancement for large multi robot systems.
- **Data analysis** for performance improvement: Plotjuggler, foxglove.
- Programming Languages: **Proficient in C++** and **experienced in Bash** scripting.
- Robot Operating System (ROS): **Strong expertise in ROS**, including the development of custom nodes, services, and integration of various ROS packages.
- **Hardware-Related Software Development:** Experience with **resource-constrained systems**, various sensors and actuators like **LIDAR, IMU, Position tracking** scanners etc.

### OBJECTIVE

As a self-taught robotics professional with a passion for cutting-edge innovation, I am looking for a challenging role that can spark my excitement for robotics. I am eager to apply my unique skillset, unconventional approach and hands-on experience with robots to develop sophisticated robotic solutions.

### WORK EXPERIENCE

#### Robotics Tech Lead

ADDVERB

2022–current

As a Tech Lead, my experience includes a wide range of responsibilities, from leading software development , supporting deployment and testing of robots to fostering growth and collaboration within the team:

- Completed end-to-end development for a new product. A mobile robot which delivers a payload of up to 40 kg at 2 m/s. My role was to support in complete development, from prototyping to dispatch and then in deployment. Developing individual device communication modules, overseeing component testing, prototype testing, software drivers, tuning drives and improving control.
- Implemented features such as dynamically changing speed and safety configurations, grid based localisation and state configuration.
- Improved dynamics during high acceleration and deceleration. This was achieved by performing detailed stability and motion analysis with different speed and load scenarios. The results were shared with mechanical design team and design changes were done in the next version of the product.
- Improved network connectivity and reliability of existing Wifi based system. Problems such as unnecessary background scan, interference due to multiple reasons, slow AP switching or no AP switching and high bandwidth consumption were sorted out. A network requirement document was formulated which is now used as a standard requirement for networking at all sites

#### Robotics Engineer

ADDVERB

2021-2022

As a Robotics Engineer, my experience includes working with guided mobile robots and contributing to various aspects of their development and deployment:

- Gained exposure to **guided mobile robots** that navigate using barcodes pasted on the ground
- Developed multiple **low-level device drivers** for navigation systems
- Supported testing and dispatch of **hundreds of robots** across different projects
- Created the robot configurator to automate validation tasks during product testing, streamlining production processes and reducing time wastage by 80%.

- Operating Systems: Expert knowledge of **Linux** and its command line interface, enabling efficient software development and debugging on robotic platforms.
- Version Control and Collaboration: Experienced with **Git**, **GitLab** (or similar), and **Docker** for collaborative software development, and **containerization**.
- **Project Management**: Demonstrated success in leading and managing projects, meeting deadlines, and ensuring customer satisfaction.
- **Leadership and Teamwork**: Proven leadership skills, with a **passion for hiring, leading, and developing people**, and a track record of working collaboratively within agile teams.
- Cross-cultural communication and collaboration skills: Having participated in **international project deployments**, I have gained valuable experience working with diverse groups of people.
- Testing and Documentation: Knowledge and motivation to write well-tested code, including unit tests and integration tests, as well as clear and concise documentation.

## EDUCATION

M.Sc. INFORMATICS

IIC, DELHI UNIVERSITY

2018-2020

B.Sc. (H) PHYSICS

DDUC, DELHI UNIVERSITY

2104-2107

## LANGUAGES

English



Hindi



- Enhanced and refined the existing code base for better performance
- Worked on a **deployment project in another country** for a major client, handling extended work shifts, manufacturing defects, and site-specific requirements
- **Improved performance and reliability of barcode reader systems** through bug fixing and the addition of sanity checks
- Refactored and rewrote the software stack for the robots, optimizing their efficiency and scalability

## GET-Mobile Robotics

Addverb

2020-2021

As a Graduate Engineer Trainee, my experience includes working on various aspects of Autonomous Mobile Robots (AMRs) that utilize natural navigation. This diverse experience has allowed me to gain expertise in several key areas:

- Gained exposure to **sensor suites for natural navigation**, including LiDAR, 3D cameras, and IMUs
- Developed proficiency in important ROS packages, such as **robot localisation**, **AMCL** and **move\_base**
- Worked on navigation and **control of AMRs**, tuning parameters of controllers like MPC, PD, and TEB to minimize deviation and maximize speed
- Implemented **path smoothing** and **velocity profiling** using parameterized curves, such as Bezier curves and cubic splines
- Supported testing, validation, and dispatch of multiple robots for different clients
- Engaged in **multi-agent path planning** using existing algorithms like AA-SIPP and continuous CBS, and customizing them for specific use cases
- Developed and tested multi-agent navigation and planning simulations using **Gazebo** and **Stage simulators**

## Mentor

Robotics Club DDUC

2017-present

As a mentor for students at my alma mater, my experience includes:

- Providing one-on-one guidance and advice to students for different robotics competitions.
- Assisting students in doing their robotics projects.
- Conducting workshops on different topics related to robotics.
- Serving as a positive role model and source of encouragement
- Facilitating connections between students and industry professionals
- Collaborating with faculty and staff to address student needs and provide comprehensive support

## TRAININGS AND CERTIFICATIONS

- **Design Patterns in C++** (online): learned about different design patterns such as singleton, factory class, etc.
- **Udacity, C++ nanodegree** (online): learned about concepts of OOP, memory management and concurrency.
- **ROS developers day**, The construct: learned about ros 2 architecture, behaviour trees and nav 2.
- Udacity, **Robotics software engineer nanodegree** (online): learned about concepts of ROS, gazebo, localization, mapping, SLAM, path planning and navigation.
- University of Pennsylvania, **robotics fundamentals** (online): learned about concepts of mathematical fundamentals for robotics, robot kinematics, kinematic path planning.
- Columbia University, **Robotics with ROS** (online): learned about different ROS concepts.